

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2022-23

Zoology

Paper: ZOB-605 — Immunology and Parasitology

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A (including Question No. 1 which is compulsory) and three questions from Section-B (including Question No. 1 which is compulsory). Use a separate answer book for each section. The figures in the right-hand margin indicate marks.

SECTION – A
(Immunology) (Marks: 35)

Question 1. Answer the following parts:

(a) State whether the following statements are True or False. Justify your answer in 2–3 sentences each: [4 × 2 = 8]

- (i) Type II hypersensitivity is mediated by immune complex.
- (ii) B-cell epitopes are always presented with MHC.
- (iii) Herd immunity is achieved in highly contagious disease when 40% of population is vaccinated.
- (iv) Graves' disease affects the thyroid gland.

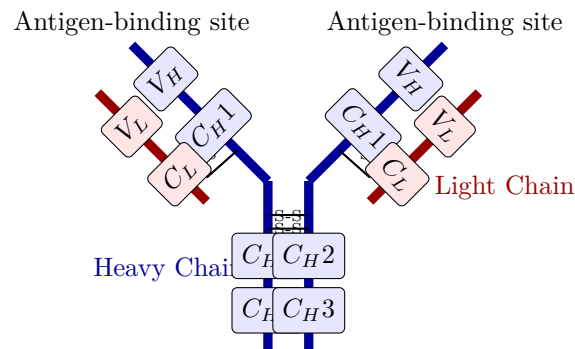
(b) Define the following: [3 × 1 = 3]

- (i) M cells
- (ii) Immunodeficiency
- (iii) Adaptive immunity.

Question 2. Describe the mechanism of antigen processing and presentation via MHC class I and class II to T lymphocytes. [12]

Question 3. Answer the following:

- (a) Discuss the properties of an ideal vaccine. [5]
- (b) Discuss the structure of immunoglobulin and describe the function of IgG, IgA and IgM. [7]



Structure of Immunoglobulin G (IgG)

Question 4. Write short notes on the following:

[3 × 4 = 12]

- (a) Lymphocytes
- (b) Hapten
- (c) Spleen.

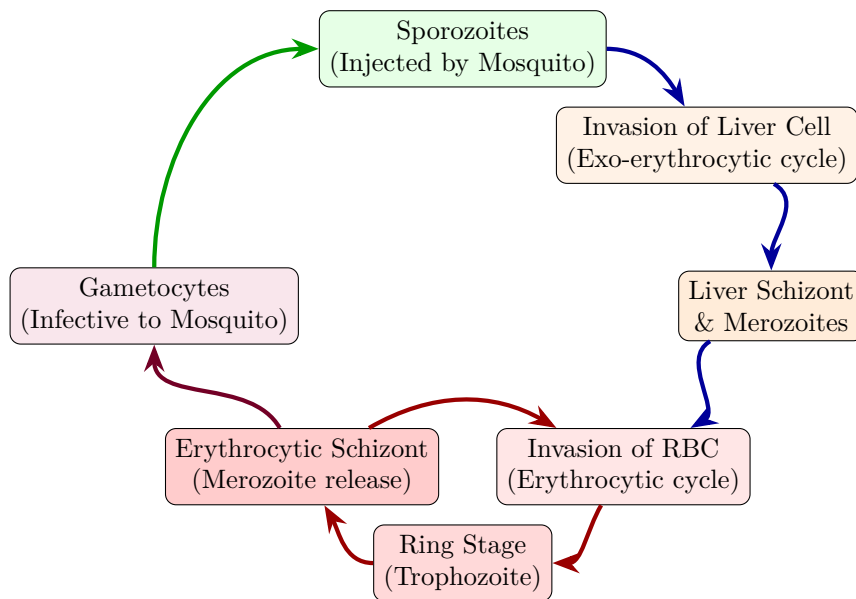
SECTION – B (Parasitology) (Marks: 35)

Question 1. Answer the following parts:

- (a) State whether the following statements are True or False. Justify your answer in 2–3 sentences each: [2 × 2 = 4]
- (i) Individual suffering from trichinosis develops myositis, fever, periorbital oedema within a week.
 - (ii) Unembryonated eggs of *Paragonimus westermani* are coughed up and voided in sputum.
- (b) Differentiate between the following: [2 × 2 = 4]
- (i) Endotoxin and Exotoxin
 - (ii) Tachyzoite and Bradyzoite.
- (c) Define the following: [3 × 1 = 3]
- (i) Hyperparasites
 - (ii) Paratenic host
 - (iii) Zooanthroponosis.

Question 2. Discuss the asexual cycle of *Plasmodium vivax*. Comment on the clinical manifestations and control measures of the disease. [12]

Asexual Life Cycle of *Plasmodium vivax*



Question 3. Describe the life cycle and pathogenesis of *Diphyllobothrium latum*. Briefly discuss its treatment and prophylactic measures. [12]

Question 4. Write notes on the following:

[2 × 6 = 12]

- (a) Host-parasite coevolution
- (b) Intestinal amoebiasis.

[Click Here](#)

[Click Here](#)

[Click Here](#)